# Solutions to Chapter 4 Exercises:

## 1. Choosing ML Models for Real-World Network Data

- Classical models are appropriate when interpretability, low latency, and small labeled datasets are priorities.
- Complex models (trees, ensembles, mixtures) are favored for nonlinear, noisy, high-dimensional data.
- Trade-offs involve explainability, computational cost, and robustness.

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## 2. Ensemble Methods in Dynamic Networks

- Ensembles improve robustness by reducing variance and combining diverse decision boundaries.
- Challenges: model disagreement, computational overhead, and stale components in fast-changing 5G/IoT/edge environments.

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## 3. Adapting Distance-Based Methods

- Use graph distances instead of Euclidean for network topologies.
- Apply time-aware metrics for temporal correlations (DTW, sliding windows).
- Respect protocol constraints by embedding routing/queuing information in feature space or distance metric.

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## 4. Interpretability vs. Predictive Power

- Logistic regression and decision trees provide interpretable decision rules essential for intrusion detection and audits.
- Ensembles achieve stronger predictive power but reduce transparency.
- Choice depends on whether correctness (interpretation) or accuracy (performance) is more critical.

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## 5. Selecting Output Layer and Loss

- Delay prediction → linear output + MSE (Gaussian).
- Binary intrusion classification → sigmoid + binary cross-entropy (Bernoulli).
- Multi-class traffic classification → softmax + categorical cross-entropy (Multinomial).

## 6. Regression Models in Networking

- Linear regression works when throughput changes proportionally with signal quality.
- Polynomial regression models nonlinear fading effects or mobility effects.
- High-degree polynomials may overfit noise due to wireless variability.

## 7. Logistic vs. Softmax vs. SVM

- Logistic regression: interpretable, simple, good for binary tasks.
- Softmax regression: natural for multi-class problems.
- SVM: effective in high-dimensional spaces but sensitive to scaling; kernel SVM is costly.

## 8. Decision Trees and Rule Extraction

- Trees generate human-readable detection rules (e.g., thresholds for packet rate).
- Compared to SVMs: more interpretable.
- Compared to ensembles: less accurate but transparent.
- Compared to GMMs: deterministic with crisp boundaries but less expressive.

## 9. Clustering Unlabeled Data

- Choose number of clusters using silhouette score, elbow method, or BIC (for GMM).
- Evaluate quality using cluster compactness and separation.
- Map clusters to meanings by inspecting dominant traffic attributes within each cluster.

## 10. k-NN Wi-Fi Signal Quality Classification

- Scaling ensures distance fairness between features.
- Small $k \rightarrow$ flexible but noisy; large $k \rightarrow$ stable but may blur class boundaries.
- Additional features (time, location) often improve separation of signal conditions.
- Reports include accuracy vs.\ $k$, confusion matrices, and decision boundary illustrations.